MIL-P-48051A (PA)
4 June 1975
SUPERSEDES
MIL-P-48051 (MU)
3 April 1972

MILITARY SPECIFICATION

PROPELLANT M10
FOR USE IN CARTRIDGES, FOR
57MM RIFLE, M18 AND M18A1

This specification is approved for use by all Departments and Agencie of the Department of Defense.

1. SCOPE

1.1 This specification covers M10 Propellant for use in the following cartridges employed with the M18 and M18A1, 57MM Recoilless Rifles:

Cartridge, HEAT, M307A1
Cartridge, WP Smoke, M308A1
Cartridge, TP, M306A1
Cartridge, HE, M306A1
Cartridge, Canister, T25E5

1.2 Classification. The propellant shall be cylindrical single perforated grains.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

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MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

DRAWINGS

ARMY

75-1-215	-	Cartridge, HEAT, M307A1, W/Fuze PI, M90A1, for 57MM Rifle
9215427	-	Cartridge, WP Smoke, M308A1, W/Fuze, PD M503A1, for 57MM Rifle
9215633	-	Cartridge, TP, M306A1, W/Fuze,PD, M503A1, for 57MM Rifle
9215030	-	Cartridge, HE, M306A1, W/Fuze, PD, Rifles, M18 and M18A1

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts:0-199

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above publication should cite: 49 CFR 0-199 (latest revision).

3. REQUIREMENTS

3.1 Materials. The propellant shall be M10, Type II (nominal web .018 inches) conforming to the requirements of Standard MIL-STD-652 except the minimum L/D ratio shall be 2.75 in lieu of 3.0. The propellant may be coated with a graphite glaze for ease of loading, but the total added graphite shall not exceed 0.2% when determined in accordance with MIL-STD-286, Method 308.1.3.

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3.2 Proving Ground Assessment. The propellant, when fired with the weight of projectile (M306A1 2.75 lb) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.4.

3.2.1 Charge Establishment. At 70 degrees Fahrenheit (°F) and standard weapon conditions (M18 Rifle), the propellant shall be capable of charge assessment to a velocity of 1,200 feet per second (fps) with a maximum average pressure of 7,800 pounds per square inch (psi) maximum and an expected recoil momentum of 3 pound seconds (lb-sec) maximum (see 6.11).

3.2.2 Uniformity of velocity. The standard deviation of velocity for the lot shall not exceed 13.0 fps.

3.2.3 Permissible Individual Maximum Pressure. The propellant at recommended charge weight shall not produce a chamber pressure greater than 9,000 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon.

3.3 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. Quality Conformance Inspection shall be conducted on this item.

4.3 Quality Conformance Inspection

4.3.1 Inspection Lot Formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078.

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4.3.2 Examination and Testing

4.3.2.1 Materials (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical physical properties in accordance with MIL-STD-652.

4.3.2.2 Proving Ground. (see 3.2) Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652.

4.3.2.2.1 Charge Establishment (see 3.2.1). The lot shall be rejected if the assessed charge weight of propellant has an expected average pressure or an expected recoil momentum which does not meet the applicable requirement.

4.3.2.2.2 Uniformity

the factor (see 6.3) times the velocity standard deviation of the rounds tested in accordance with 4.4.1.2 exceeds the applicable requirement the lot shall be rejected.

4.3.2.2.2.2 Pressure (see 3.2.3)

4.3.2.2.2.2.1 Initial Production. If any individual pressure observed during uniformity firings at any operating temperature after correction for difference between as fired and recommended service charge weight exceeds 9,000 psi, the lot shall be rejected.

4.3.2.2.2.2.2 Subsequent Production. If any individual pressure at 70°F, when corrected to standard conditions and recommended charge weight, exceeds 9,000 psi minus the temperature/pressure coefficient times 75°F, the lot shall be tested at 145°F. If any individual pressure at 145°F, when corrected to standard conditions and recommended charge weight, exceeds 9,000 psi, the lot shall be rejected.

4.4 Test methods and procedures

4.4.1 Proving ground assessments. These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

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4.4.1.1 Charge Establishment

4.4.1.1.1 Charge Slope Method. Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2.1). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.4.1.2). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. If the final adjustment in charge weight exceeds 2 percent of the weight used in uniformity firings, the lot shall be retested for charge establishment (see 6.9).

4.4.1.1.2 Constant Slope Method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite velocity versus charge weight versus pressure curve can be established from test results generated. This curve can then be used as in 4.4.1.1.1 above to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method. If the final charge weight adjustment exceeds 2 percent of the weight used in uniformity firings, the lot shall be retested for charge establishment in accordance with 4.4.1.1.1 (see 6.9).

4.4.1.2 Uniformity

4.4.1.2.1 Initial Production. Thirty rounds shall be assembled with the established charge weight of propellant. Ten rounds each shall be conditioned to -50°F, 70°F and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions).

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Firing results at all temperatures shall be corrected to standard projectile weight (2.75 lb) and to recommended charge weight conditions (i.e. the difference between as fired charge weight and recommended charge weight). The means and standard deviation of the velocities, pressures and recoil momentums at all firing temperatures shall be calculated using corrected results. The average pressure-temperature coefficient for the lot shall be calculated from the pressure data obtained at 70°F and 145°F. The average of the coefficients for 10 lots shall be utilized to calculate a 70°F upper pressure control for use in subsequent production.

4.4.1.2.2 Subsequent Production. Ten (10) rounds shall be assembled and tested at 70°F, fired alternately with calibration rounds. The means and standard deviations of the velocity, pressure and recoil momentums shall be calculated on results corrected to standard projectile weight (2.75 lb), standard weapon/firing conditions and recommended charge weight conditions (see 4.3.2.2.2.2.2). If any individual pressure exceeds the calculated 70°F upper pressure control (see 4.4.1.2.1), ten additional test rounds shall be conditioned and fired at 145°F. All results at 145°F shall be corrected to standard projectile weight (2.75 lb) and recommended charge weight conditions. If any individual round at 145°F exceeds 9,000 psi, the lot shall be rejected.

4.4.1.3 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested. A lot shall be retested whenever any of the following conditions exist and whenever the proof director deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds two percent of the expected muzzle velocity.

c. If calibration components are available, but not utilized in either test or calibration rounds (see 6.9)

d. Any test round result is an outlier (see 6.10)

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4.4.1.4 Referral. A test lot shall be referred to Picatinny Arsenal, ATTN: SARPA-QA-A-P for disposition whenever any of the following conditions exist or whenever the Proof Director deems it necessary:

a. Number of test or calibration rounds and/or results utilized is less than specified.

b. Calibration components are not available.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652.

6. NOTES

6.1 Ordering Data. Ordering data shall be in accordance with MIL-A-48078(PA) with the following exceptions.

6.1.1 Procurement Requirements. Procurement requirements of MIL-A-48078(PA) shall apply except for First Article Sample.

6.1.2 Contract Data Requirements. In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data and one copy of the Propellant Description sheet shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-P, and one copy to ATTN: SARPA-AD-E-P-2, Dover, New Jersey 07801.

6.2 Standard deviation. The standard deviation shall be calculated with (n-1) as the devisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

6.3 Factors. The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.4 Initial production lots. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Lapse in production of 90 days or more shall cause testing to revert back to initial production plan.

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6.5 Tolerance. A tolerance of -5°F applies to all -50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

6.6 Surveillance sample. Samples shall be forwarded to Picatinny Arsenal, Dover, New Jersey, ATTN: SARPA-FR-G for the propellant surveillance test program in accordance with MIL-STD-652.

6.7 Test Charges. Test charges shall be assembled in accordance with drawings: 9215633 or drawing 9215030.

6.8 Proving ground test summary

Initial Production

Temp (°F)	Sample Size	Vel(fps) (assessed)	Lot Std. Dev of Vel (fps)	Perm Ind. Max Pres (psi)	Avg. Pres. Max (psi)
-50	10	N/A	N/A	9,000	N/A
70	10	1200	13.0	9,000	7,800
145	10	N/A	N/A	9,000	N/A

Subsequent Production

70	10	1200	13.0	9,000	7,800
*145	10	N/A	N/A	9,000	N/A

*To be fired only when 70°F results indicate thar marginal pressure performance may exist (see 4.3.2.2.2.2.2).

6.9 Retest. All original and retest data shall be submitted to Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801.

6.10 Outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_t) as follows:

$$T_T = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

Where:

X_1 = test result being tested as outlier
X_2 = test results adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test results adjacent to the largest (or smallest) test results

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X_1 is an outlier if T_t is greater than T_c.

Figure 1 presents a schematic of the decision process imposed when the outlier test is utilized.

6.11 Expected Pressure. If the expected pressure for a lot of propellant is less than 5,900 psi or greater than 6,700 psi, the lot shall be restricted from use in the T25E5 cartridge.

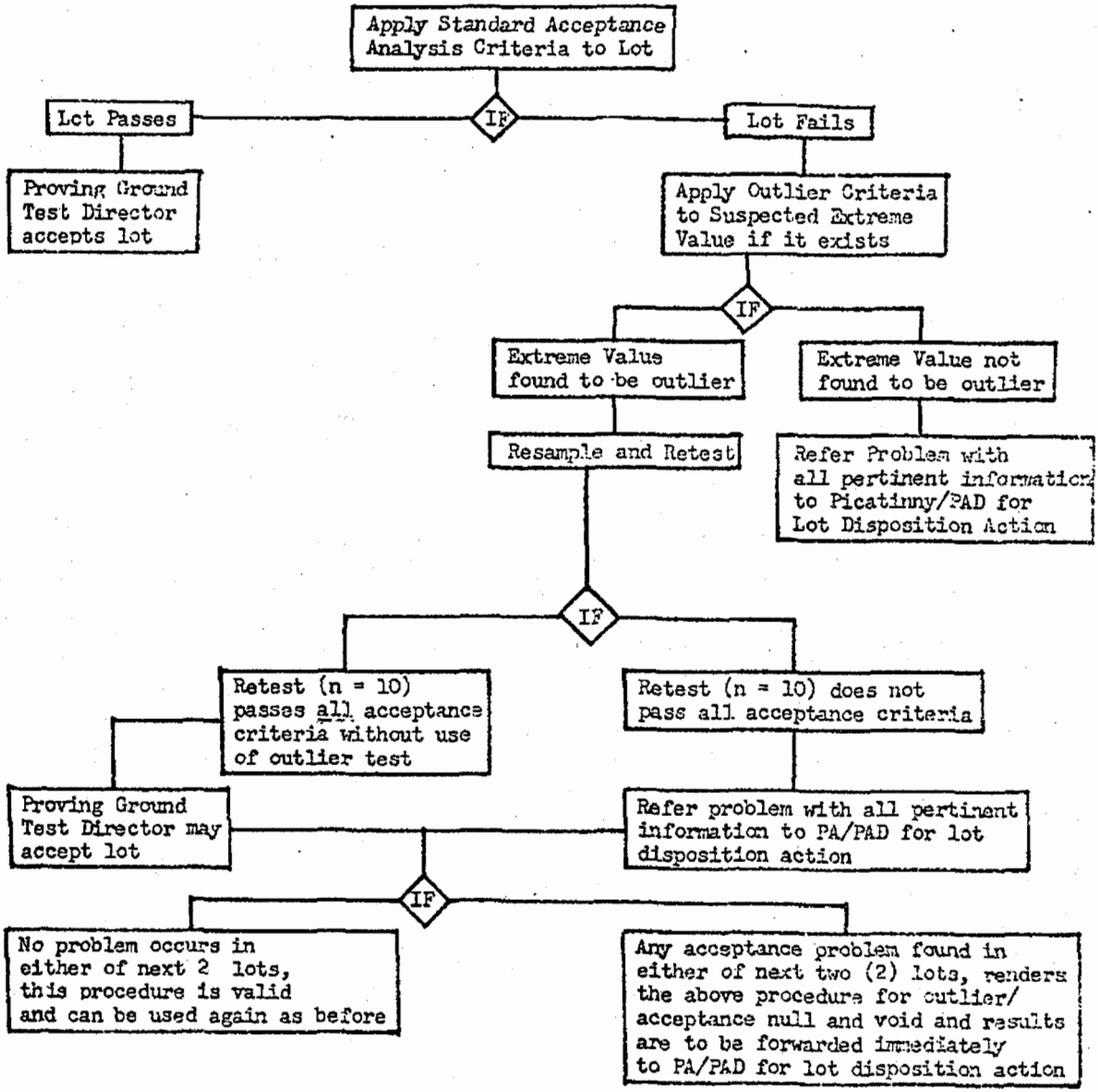
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FIGURE 1. DECISION PROCESS WITH OUTLIER TEST



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